

HIGH-LEVEL DESIGN (HLD)

RAG-Based Customer Support Assistant with LangGraph & HITL

1. System Overview

◆ Problem Definition

Traditional customer support systems rely heavily on human agents, leading to:

- Delayed responses
 - High operational costs
 - Inconsistent answers
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◆ Proposed Solution

The system is a Retrieval-Augmented Generation (RAG) based Customer Support Assistant that:

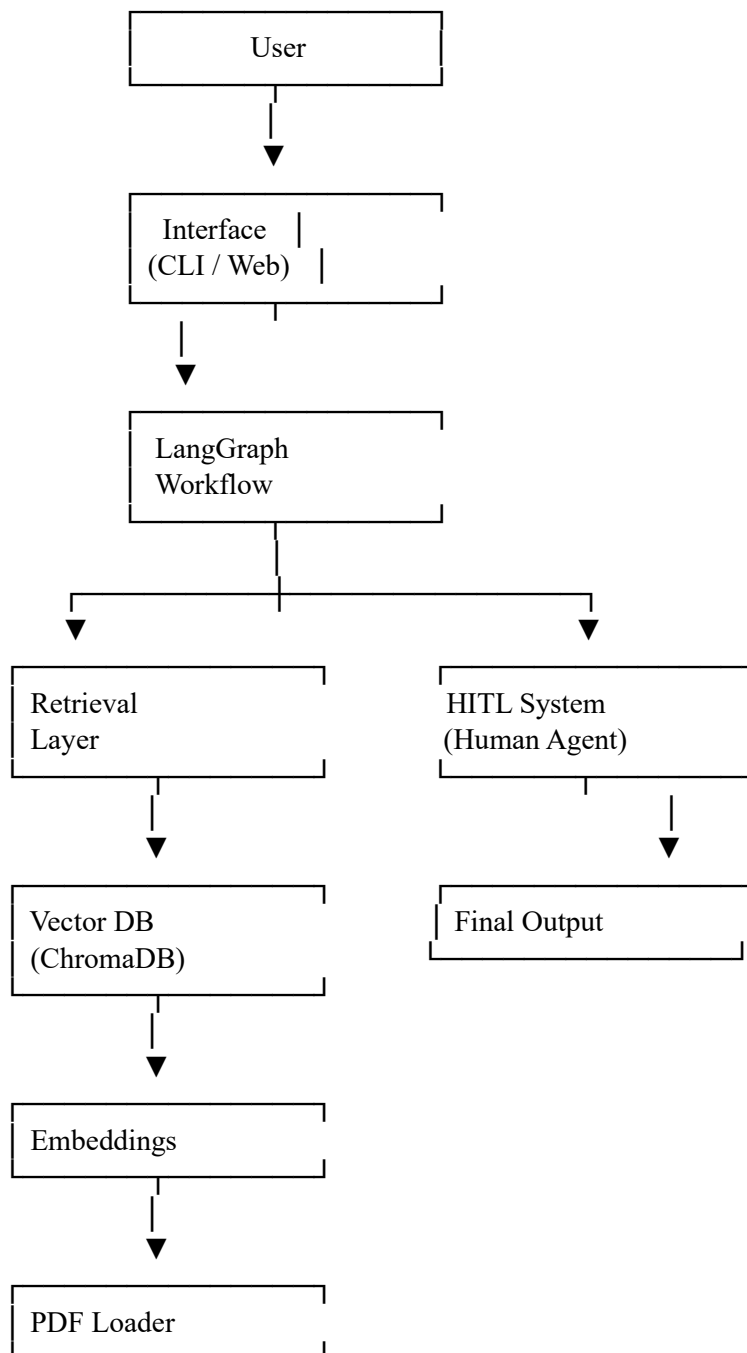
- Processes a PDF knowledge base
 - Retrieves relevant information using embeddings
 - Generates contextual answers using an LLM
 - Uses LangGraph workflow for control logic
 - Supports Human-in-the-Loop (HITL) escalation
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◆ Scope of the System

The system:

- Answers queries using document knowledge
 - Handles unknown queries via escalation
 - Provides a scalable AI-based support solution
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2. Architecture Diagram (Mandatory)



3. Component Description

- ◆ Document Loader

Loads PDF using PyPDFLoader and extracts text content.

- ◆ Chunking Strategy

Splits document into smaller chunks using:

- Chunk Size: 500
 - Overlap: 50
Ensures better context retention.
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◆ Embedding Model

Uses HuggingFace model (all-MiniLM-L6-v2) to convert text into vectors.

◆ Vector Store

ChromaDB is used to:

- Store embeddings
 - Perform similarity search
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◆ Retriever

Fetches relevant chunks based on query similarity.

◆ LLM (Language Model)

Generates final answers using retrieved context.

◆ Graph Workflow Engine

Implemented using LangGraph:

- Controls execution flow
 - Enables modular design
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◆ Routing Layer

Decides whether:

- Return answer
 - OR escalate to human
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◆ HITL Module

Handles:

- Complex queries
- Low-confidence responses

- Missing context

4. Data Flow

◆ Document Processing Flow

PDF → Loader → Chunking → Embeddings → ChromaDB

◆ Query Processing Flow

User Query → LangGraph → Retriever → Context



LLM → Generate Answer



Decision (Routing)



Output OR HITL

5. Technology Choices

◆ ChromaDB

- Lightweight
 - Fast
 - Easy to integrate
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◆ LangGraph

- Supports workflow-based AI systems
 - Enables conditional routing
 - Ideal for HITL
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◆ Embedding Model

- Free and efficient
 - Good semantic understanding
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◆ LLM Choice

- Flexible (OpenAI / HuggingFace)

- Generates contextual responses
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6. Scalability Considerations

◆ Large Documents

- Efficient chunking
 - Vector indexing
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◆ High Query Load

- Use caching
 - Optimize retrieval
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◆ Latency Optimization

- Reduce chunk size
 - Use lightweight models
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◆ Future Scalability

- Multi-document support
 - API deployment
 - Distributed vector databases
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7. Summary

This system demonstrates a scalable and modular RAG-based architecture integrated with:

- Workflow orchestration (LangGraph)
- Conditional routing
- Human-in-the-Loop support

It provides a robust solution for automating customer support while maintaining human fallback capabilities.